Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_______________show ip route ospf____

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _S0/0/0___

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _129__

Does R2 show up as an OSPF neighbor on R1? __yes__

Does R2 show up as an OSPF neighbor on R3? __no___

What does this information tell you?

____When an interface is set to passive, it stops sending and receiving OSPF Hello packets, so no neighbor adjacency is formed on that interface. R3 can still reach 192.168.2.0/24 but only through R1 as an intermediary, resulting in a higher cost path. R2 remains a neighbor of R1 because R2's S0/0/0 (facing R1) is still active._______

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_________R2(config)

R2(config-router) __________

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/1__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_____cost(R3 S0/0/1) = 64 + cost(R2 G0/0) = 1 = 65_______________

Is R2 listed as an OSPF neighbor to R3? __yes__

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

____The default reference bandwidth is 100 Mbps. Modern networks use Gigabit (1 Gbps) and 10 Gigabit links, which would all receive the same OSPF cost of 1 if the default is kept, making it impossible to differentiate between them. Increasing the reference bandwidth (e.g., to 1000 or 10000) allows OSPF to assign more granular and accurate costs to high-speed interfaces, enabling better path selection.______

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Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

__192.168.3.0/24 from R1: _R1 S0/0/1 cost = 100,000 / 128 = 781_________R3 G0/0 cost = 100,000 / 100,000 = 1___ Total = 78__

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______ 192.168.23.0/30 from R1: __ 781 (R1 serial) + 781 (R3 or R2 serial) = 1562_____

____

1

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

______ All serial interfaces now have bandwidth 128 Kbps. The path from R1 to 192.168.23.0/30 traverses two serial links: cost = 100,000/128 + 100,000/128 = 781 + 781 = 1562._____________________

2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

____ After setting ip ospf cost 1565 on R1's S0/0/1, the direct path to R3 has cost = 1565 + 1 = 1566. The alternative path via R2 (R1 S0/0/0 → R2 S0/0/1 → R3 G0/0) has cost = 781 + 781 + 1 = 1563. Since 1563 < 1566, OSPF selects the lower-cost path through R2, as confirmed by O 192.168.3.0/24 [110/1563] via 192.168.12.2 in the routing table.___

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   ___ Without explicit control, the router ID is derived from the highest active IP address, which may change if interfaces are added or removed. A changing router ID causes OSPF to reset adjacencies and rebuild the LSDB, temporarily disrupting routing. Manually assigning stable router IDs (via router-id command or loopback) ensures consistent identification, predictable neighbor relationships, and stable DR/BDR elections on broadcast networks____

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2. Why is the DR/BDR election process not a concern in this lab?

   ____ DR/BDR election only occurs on multi-access broadcast networks (e.g., Ethernet with multiple routers). All inter-router links in this lab are serial point-to-point connections, confirmed by the POINT_TO_POINT state in show ip ospf interface output. Point-to-point links have only two endpoints, so no DR/BDR election is needed.____

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3. Why would you want to set an OSPF interface to passive?

   ___ Passive interfaces stop sending OSPF Hello packets, which prevents unnecessary routing traffic on LAN segments where only end hosts (PCs) are connected—no other OSPF routers exist there. It also improves security by preventing rogue routers from forming unauthorized adjacencies. Importantly, the connected network is still advertised to the rest of the OSPF domain; only neighbor discovery is suppressed._____

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   ____ The use of multiple areas with Area 0 (backbone) as the hierarchy center. Dividing a large network into areas confines LSA flooding within each area's boundary, reduces each router's LSDB size, and limits the impact of topology changes to only the affected area. This allows OSPF networks to scale to thousands of routers without overwhelming every router with full topology information.___

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   ____ network 10.0.0.0 0.255.255.255 area 0__________

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   ___ The routing decision is based on Administrative Distance (AD), not the protocol's own metric. AD values are: EIGRP = 90, OSPF = 110, RIP = 120. The router selects the route with the lowest AD, so the EIGRP route (AD 90) is chosen, regardless of its larger metric number.____

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